

01 / APPROACH

Predicting freight rates for future loads

Machine Learning Engineer Assessment

A documented workflow from raw inputs to reproducible predictions.

\$637.66

Primary model holdout RMSE

\$122.33

Primary model holdout MAE

5.11%

Primary model holdout WAPE

Decision

Use `catboost_full_noquote_log_rpm` for the 12,000 November-December loads, then use `catboost_reduced_log_rpm` for the fixed December scenario whose input schema has fewer fields.

Choose model configurations on a July-August temporal selection window, evaluate the frozen choice on September-October, and refit the two selected configurations on all January-October labeled rows for submission.

What this submission demonstrates

The dataset calls for future-period validation: the labeled development data ends in October and the final labels are withheld for November-December.

Weights contain missing and nonpositive values, new cities appear in validation, and the quote signal changes its relationship with price between development months. These are explicit pipeline and feature-policy decisions.

The primary model lowers holdout RMSE by 4.9% versus the distance/equipment baseline. This is an internal retrospective result, not the organizer's unseen validation score.

Evidence boundary

The supplied scorer checks IDs, schema, finite positive predictions and the December grid. It does not calculate accuracy. November-December prediction accuracy remains unknown until the organizer scores the withheld labels.

02 / DATA

Data scope and preparation

Audit all supplied files without modifying them. Preserve the identifier and input schemas, and keep source hashes so the results can be tied to the exact inputs.

Input	Rows	Dates	Role
train_test.csv	48,000	2025-01-01 to 2025-10-31	Labeled development data
validation.csv	12,000	2025-11-01 to 2025-12-31	Final prediction only; no labels
December chart inputs	31	2025-12-01 to 2025-12-31	Fixed lane scenario; fewer features

Data-quality checks and treatment

Issue	Train / validation	Treatment or implication
Missing weight	300 / 165	Keep a missing/invalid flag; impute numeric values using training-only medians.
Nonpositive weight	292 / 145	Treat as missing. Do not take absolute values or discard the full load.
Missing market_index	374 / 249	Training-only numeric imputation plus a missingness indicator in the primary model.
Duplicate identifiers / rows	0 / 0 extra ID rows; 0 / 0 exact duplicate rows	Keep every supplied load; preserve the template's order when writing predictions.
Unseen locations and lanes	8 new cities; 1,447 affected rows	1,461 validation loads also have an unseen directed lane. Support unseen categorical values; the primary model retains supplied geography.
Rare expensive targets	Maximum \$25,533.00	Retain labeled rates: size alone does not prove an error. Compare losses/target transforms and inspect error concentration.

The audit finds no numeric parse failures, nonfinite numeric inputs, invalid ISO dates, overlapping load IDs or exact duplicate rows. Coordinate checks verify mathematical bounds only; geographic correctness is unverified.

Counts above describe source inputs. Data-quality transformations are fitted independently inside each training partition, then reused unchanged on its later evaluation partition.

03 / FEATURES

A reproducible feature pipeline

One deterministic feature function is shared by training and inference. `load_id` is used only to align outputs; `posted_rate` is the target and never enters the feature matrix.

Group	Derived or retained features	Availability
Shipment / lane	Distance, log/square-root distance, cleaned weight, weight missingness; pickup, delivery, equipment and directed lane	Primary and reduced models
Calendar	Day of week/month/year, days from a fixed origin, weekend flag; weekly, annual and monthly sine/cosine terms	Derived only from supplied date
Geography	Supplied pickup/delivery coordinates and coordinate differences	Primary model only
Market	<code>market_index</code> and its missingness flag	Primary model only
Quote signal	Excluded from both submission models; quote-based candidates remain diagnostic benchmarks	Provenance and stability unresolved
Missing values	Categorical missing token; training-only numeric medians; no future-data lookup table	Fitted on training partition only

Why this model

CatBoost learns nonlinear shipment, calendar and location interactions and supports unseen categories. Ridge provides a regularized comparator with one-hot categories and distance interactions.

The model fits $\log(\text{posted_rate} / \text{distance})$, then exponentiates and multiplies by the supplied distance. This keeps the rate-per-mile structure while selection uses RMSE in dollars.

Fixed configuration: 700 iterations, depth 6, learning rate 0.055, seed 42. Select by dollar RMSE; the later holdout does not tune features, loss or stopping.

Feature availability and leakage

January-August `quote_signal` associations reverse sign and become nearly zero in August. Exclude this unstable field conservatively; the observations do not prove target leakage. Observed rate per mile remains a training target/diagnostic only.

Operational assumption

Signal definitions and timestamps are unspecified. The primary model assumes `market_index` is available before pricing. Verify its calculation and timing before operational use.

04 / VALIDATION

Chronological validation

Preserve time order to match the organizer's future-period task. Keep September-October separate from candidate comparison, then evaluate the frozen selections once on that later window.

Stage	Training data	Later data / use
Candidate selection	Jan-Jun 2025: 28,806 rows	Jul-Aug 2025: 9,671 rows; select models
Locked holdout	Jan-Aug 2025: 38,477 rows	Sep-Oct 2025: 9,523 rows; report accuracy
Final refit	Jan-Oct 2025: 48,000 rows	Nov-Dec: 12,000 unlabelled rows; no tuning

Model comparison

Model	Evaluation window	RMSE (\$)	MAE (\$)	WAPE
Primary selected model	Jul-Aug selection	644.54	206.04	8.67%
Reduced-input model	Jul-Aug selection	643.92	202.17	8.50%
Distance/equipment baseline	Sep-Oct holdout	670.66	229.10	9.58%
Primary selected model	Sep-Oct holdout	637.66	122.33	5.11%
Reduced-input model	Sep-Oct holdout	637.61	124.13	5.19%

Primary configuration: catboost_full_noquote_log_rpm. Reduced configuration: catboost_reduced_log_rpm. All candidate results and configurations are retained in artifacts/metrics.json and experiments/results.json.

RMSE = sqrt(mean squared dollar error); MAE = mean absolute dollar error; WAPE = sum absolute error / sum observed rates. Selection and holdout results use different training histories and are not interchangeable.

Primary holdout R-squared is 0.8254. This summarizes the supplied development period; it cannot establish performance on new cities or market conditions in November-December.

Controlled cold-city stress

Controlled cold-city stress: deterministically select 8 January-June cities without labels, remove all training loads touching them, and refit the already selected primary configuration on 22,894 retained rows.

Evaluate 2,004 July-August loads touching those cities: RMSE \$540.54, MAE \$141.82, WAPE 7.04%. This post-selection test does not tune the model and does not measure the actual November-December unseen cities.

05 / DECEMBER

A fixed lane, 31 future dates

Predict 31 daily loads from Lexington to Fort Wayne, keeping distance at 360 miles, equipment as Dry Van and weight at 32,000 lb. Only the supplied date varies.

Candidate: December 2025 Predicted Load Rate

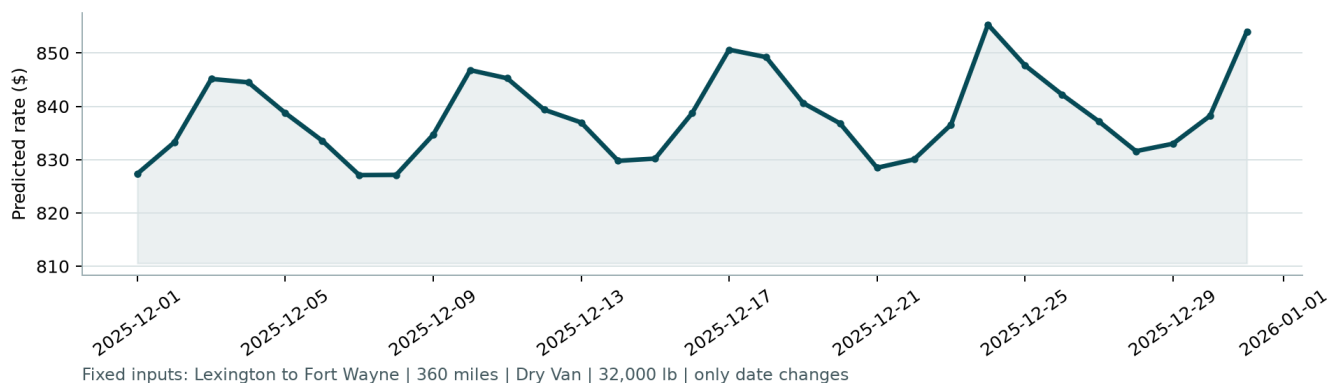


Figure: candidate_december.png generated by the organizer's unmodified score.py from the completed December CSV.

Handling the reduced input contract

The scenario has no coordinates, market_index or quote_signal. Use the separately selected catboost_reduced_log_rpm, which is trained and evaluated using the same six base input columns, rather than supplying invented future signals.

The reduced model shares shipment, lane and calendar transformations with the primary model. It needs no external city lookup, forecast market index or future quote. Refit it on all January-October labeled rows before producing December predictions.

The reduced model's September-October RMSE is \$637.61. That result measures its complete reduced feature contract across the held-out loads, not accuracy for this particular December lane.

Interpretation

This is a model scenario, not an observed December price curve. Daily shape reflects learned calendar effects under fixed inputs. One year of January-October history cannot establish a repeated December seasonal pattern, and market changes remain unobserved here.

06 / REPRODUCIBILITY

Run, inspect and submit

The repository contains the feature/model code, fixed development protocol, recorded metrics, final predictions and the report source. Follow README.md to reproduce the submission; the candidate records and submits their own Loom walkthrough.

Artifact	Purpose
validation_predictions.csv	Exactly load_id,predicted_rate; 12,000 loads aligned to the supplied template
submission/december_chart_predictions.csv	31 completed daily rows; original seven-column schema preserved
artifacts/full_model.joblib / reduced_model.joblib	Final refitted models and training-only preprocessing state
artifacts/metrics.json	Candidate selection, holdout scores, configuration and reproduction metadata
reports/data_audit.json	Input hashes, quality counts and distribution checks
reports/freight_rate_report.pdf	This report, including the complete official December chart

Reproduction

```
python -m pip install -r requirements.txt
```

```
python -m src.run_all
```

```
python -m src.train --recompute-selection
```

The default complete workflow audits inputs, trains/refits selected configurations, writes both prediction files, runs the original scorer and regenerates this PDF. Recompute selection to independently rerun the recorded candidate comparison.

Official scorer acceptance establishes file structure and finite positive predictions. Meaningful pipeline tests additionally cover input sanitization, target/ID exclusion, missing values and the reduced-input schema.

Limitations and follow-up

Future labels are unavailable. Temporal holdout is a defensible retrospective estimate, but final period performance is still unknown.

All holdout cities are represented in its training history. A separate controlled cold-city stress provides supporting evidence, without establishing accuracy on the actual new validation cities.

Rare expensive loads can dominate dollar RMSE. Preserve them, inspect subgroup errors and validate unusual source prices before deciding on any future removal or correction policy.

Before operational use, verify signal timestamps, collect additional seasonal cycles and backtest on several rolling origin windows, including truly unseen locations.

Evidence: supplied assessment PDF/README and unchanged scorer; recorded read-only data audit; recorded July-August candidate results and September-October holdout results. Input/model hashes, package versions, random seed and thread count are recorded in artifacts/metrics.json.